Chapter-7 Solution of Triangle

- A triangle ABC consists of six elements : 3 sides (i.e. a, b and c) and 3 angles (i.e. A, B and C).

- The process of finding the unknown elements of a triangle from its element is known as ~~solution of~~ Solving the triangle.

Exercise-7

① In $\triangle ABC$, $B = 60°$, $b : c = \sqrt{3} : \sqrt{2}$ Show that $A = 75°$.

→ Soln,

$$\frac{b}{c} = \frac{\sqrt{3}}{\sqrt{2}}$$

or, $\frac{\cancel{2R}\sin B}{\cancel{2R}\sin C} = \frac{\sqrt{3}}{\sqrt{2}}$

or, $\frac{\sin 60°}{\sin C} = \frac{\sqrt{3}}{\sqrt{2}}$

or, $\frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{\sqrt{2}} \times \sin C$

or, $\frac{\cancel{\sqrt{3}}}{\cancel{\sqrt{2}}\cdot\sqrt{2}\cdot\cancel{\sqrt{3}}} \times \cancel{\sqrt{2}} = \sin C$

or, $\sin C = \frac{1}{\sqrt{2}}$

or, $C = 45°$

We know that,

$A + B + C = 180°$

or, $A + 60° + 45° = 180°$

or, $A = 180° - 105°$

$\therefore A = 75°$

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② If $\cos A = \frac{4}{5}$, $\cos B = \frac{3}{5}$, find $a:b:c$

→ Soln,

$\sin A = \sqrt{1-\cos^2 A}$

$= \sqrt{1-\frac{4^2}{5^2}} = \sqrt{1-\frac{16}{25}} = \sqrt{\frac{9}{25}}$

$= \frac{3}{5}$

$\sin B = \sqrt{1-\frac{3^2}{5^2}} = \sqrt{\frac{16}{25}} = \frac{4}{5}$

And,

$\sin C = \sin(A+B)$

$= \sin A \cdot \cos B + \cos A \cdot \sin B$

$= \frac{3}{5} \times \frac{3}{5} + \frac{4}{5} \times \frac{4}{5}$

$= \frac{9}{25} + \frac{16}{25}$

$= \frac{25}{25}$

$= 1$

Now,

$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$

or, $\frac{a}{3/5} = \frac{b}{4/5} = \frac{c}{1}$

or, $\frac{a}{3} = \frac{b}{4} = \frac{c}{5}$

$\therefore a:b:c = 3:4:5$ //.

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③ If three angles of a triangle are in the ratio 2:3:7, find a:b:c.

↳ Soln,

~~Let~~ $A:B:C = 2:3:7$

let, $A = 2x$, $B = 3x$ and $C = 7x$.

Now,

$A + B + C = 180°$

or, $2x + 3x + 7x = 180°$

or, $12x = 180°$

or, $x = \frac{180}{12}$

$\therefore x = 15°$

so,

$A = 2x = 2 \times 15 = 30°$

$B = 3x = 3 \times 15 = 45°$

$C = 7x = 7 \times 15 = 105°$

~~At~~ Last,

Using sine law,

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

or, $$\frac{a}{\sin 30°} = \frac{b}{\sin 45°} = \frac{c}{\sin 105°}$$

or, $$\frac{a}{1/2} = \frac{b}{1/\sqrt{2}} = \frac{c}{\frac{\sqrt{3}+1}{2\sqrt{2}}}$$

or, $$\frac{a}{\frac{1}{\cancel{2}} \times \cancel{2}\sqrt{2}} = \frac{b}{\frac{1}{\cancel{\sqrt{2}}} \times 2\cancel{\sqrt{2}}} = \frac{c}{\frac{\sqrt{3}+1}{\cancel{2\sqrt{2}}} \times \cancel{2\sqrt{2}}}$$

or, $$\frac{a}{\sqrt{2}} = \frac{b}{2} = \frac{c}{\sqrt{3}+1}$$

$\therefore a:b:c = \sqrt{2} : 2 : (\sqrt{3}+1)$

5. Solve that $\triangle ABC$, if $A=60°$, $B=45°$ and $c=6\sqrt{2}$

→ Soln,

$A+B+C=180°$

or, $60°+45°+C=180°$

$\therefore C=75°$

by sine law,

$$\frac{a}{\sin A}=\frac{b}{\sin B}=\frac{c}{\sin C}$$

or, $\frac{a}{\sin 60°}=\frac{b}{\sin 45°}=\frac{6\sqrt{2}}{\sin 75°}$

or, $\frac{a}{\sqrt{3}/2}=\frac{b}{1/\sqrt{2}}=\frac{6\sqrt{2}}{\frac{\sqrt{3}+1}{2\sqrt{2}}}$

or, $\frac{a}{\sqrt{3}/\cancel{2}\times\cancel{2}\sqrt{2}}=\frac{b}{\frac{1}{\cancel{\sqrt{2}}}\times 2\cancel{\sqrt{2}}}=\frac{6\sqrt{2}}{\frac{\sqrt{3}+1}{2\cancel{\sqrt{2}}}\times 2\cancel{\sqrt{2}}}$

or, $\frac{a}{\sqrt{6}}=\frac{b}{2}=\frac{6\sqrt{2}}{\sqrt{3}+1}$

From first and last ratio,

$\frac{a}{\sqrt{6}}=\frac{6\sqrt{2}}{\sqrt{3}+1}$

or, $a=\frac{6\sqrt{12}}{\sqrt{3}+1}$

From last two ratios,

$\frac{b}{2}=\frac{6\sqrt{2}}{\sqrt{3}+1}$

$b=\frac{12\sqrt{2}}{\sqrt{3}+1}$

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u) If $a:b:c = 4:5:6$ in ΔABC, prove that $C = 2A$.

Soln,

~~Sin~~ Using sine law,

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

From cosine law,

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

$$= \frac{(5x)^2 + (6x)^2 - (4x)^2}{2 \times 5x \times 6x}$$

$$= \frac{3}{4}$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab} = \frac{(4x)^2 + (5x)^2 - (6x)^2}{2 \times 4x \cdot 5x}$$

$$= \frac{1}{8}$$

Now,

$$\cos 2A = 2\cos^2 A - 1 = 2 \times \left(\frac{3}{4}\right)^2 - 1$$

$$= 2 \times \frac{9}{16} - 1$$

$$= \frac{1}{8}$$

$$\cos C = \cos 2A = \frac{1}{8}$$

$\therefore C = 2A$

proved

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6. a) $a = \sqrt{3}+1$, $b = \sqrt{3}-1$ and $c = 60°$.

↳ Soln,

By cosine law,

$$\cos C = \frac{a^2+b^2-c^2}{2ab}$$

$$\cos 60 = \frac{(\sqrt{3}+1)^2 + (\sqrt{3}-1)^2 - c^2}{2.(\sqrt{3}+1)(\sqrt{3}-1)}$$

or, $\frac{1}{2} = \frac{3 + \cancel{2\sqrt{3}} + 1 + 3 - \cancel{2\sqrt{3}} + 1 - c^2}{2(3-1)}$

or, $\frac{1}{\cancel{2}} = \frac{8-c^2}{\cancel{4}\,2}$

or, $8 - c^2 = 2$

or, $c^2 = 8-2$

or, $c = \sqrt{6}$

Now,

Using cosine law,

$$\cos A = \frac{b^2+c^2-a^2}{2bc}$$

$$= \frac{(\sqrt{3}-1)^2 + (\sqrt{6})^2 - (\sqrt{3}+1)^2}{2(\sqrt{3}-1)\times\sqrt{6}}$$

$$= \frac{\cancel{3} - 2\sqrt{3} + \cancel{1} + 6 - \cancel{3} - 2\sqrt{3} - \cancel{1}}{}$$

$$= \frac{6-4\sqrt{3}}{2(\sqrt{3}-1)\sqrt{6})}$$

$A = 105°$

Again,

we have,

$A+B+C = 180°$

or, $105 + B + 60° = 180°$

or, $B = 180° - 165°$

$\therefore B = 15°$

$\therefore c = \sqrt{6}$, $A = 105°$, $B = 15°$

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7 b) $a=2$, $b=\sqrt{6}$ and $c=\sqrt{3}-1$

4 Soln,

By cosine law,

$$\cos A = \frac{b^2+c^2-a^2}{2bc}$$

$$= \frac{(\sqrt{6})^2+(\sqrt{3}-1)^2-2^2}{2\cdot\sqrt{6}\cdot(\sqrt{3}-1)}$$

$$= \frac{6+3-2\sqrt{3}+1-4}{2\sqrt{6}(\sqrt{3}-1)}$$

$$= \frac{6-2\sqrt{3}}{2\sqrt{6}(\sqrt{3}-1)}$$

$$= \frac{\cancel{2}\sqrt{3}\,\cancel{(\sqrt{3}-1)}}{\cancel{2}\sqrt{6}\,\cancel{(\sqrt{3}-1)}}$$

$$= \frac{1}{\sqrt{2}}$$

$\therefore A = 45°$

$$\cos C = \frac{a^2+b^2-c^2}{2ab}$$

$$= \frac{2^2+(\sqrt{6})^2-(\sqrt{3}-1)^2}{2\times2\times\sqrt{6}}$$

$$= \frac{4+6-3+2\sqrt{3}-1}{4\sqrt{6}}$$

$$= \frac{6+2\sqrt{3}}{4\sqrt{6}}$$

$\therefore C = 15°$

Now,

$A+B+C=180°$

or, $45°+B+15°=180°$

or, $B=180°-60$

$B=120°$

$\therefore B=120°$ #,

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9. Find the ambiguous solution of $\triangle ABC$ if.

a) $a = 2, b = \sqrt{3} + 1, A = 45°$

→ By sine law,

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

or, $$\frac{2}{\sin 45} = \frac{\sqrt{3}+1}{\sin B} = \frac{c}{\sin C}$$

or, $$2\sqrt{2} = \frac{\sqrt{3}+1}{\sin B} = \frac{c}{\sin C}$$

From 1st ~~2~~ two ratios,

$$\sin B = \frac{\sqrt{3}+1}{2\sqrt{2}}$$

$\therefore B = 75°$ or $105°$

Case I - Suppose $B = 75°$

$A + B + C = 180°$

or, $45° + 75° + C = 180°$

or, $C = 60°$

$\therefore C = 60°$

~~Case~~ $$2\sqrt{2} = \frac{c}{\sin 60°}$$

or, $$c = \cancel{2}\sqrt{2} \times \frac{\sqrt{3}}{\cancel{2}}$$

or, $c = \sqrt{6}$

Case II - Suppose $B = 105°$

$A + B + C = 180°$

or, $45° + 105° + C = 180°$

or, $C = 180° - 150°$

$\therefore C = 30°$

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Now,

$$2\sqrt{3} = \frac{c}{\sin 30}$$

or, $c = \cancel{2}\sqrt{3} \times \frac{1}{\cancel{2}}$

or, $c = \sqrt{3}$

$\therefore$ $B = 75°$, $C = 60°$, $c = \sqrt{6}$ or,
$B = 75°$, $C = 105°$, $c = \sqrt{3}$ ✓.

6. b) $a = 2$, $b = \sqrt{3}+1$ and $C = 60°$

↳ Soln,

Using cosine law,

$$\cos C = \frac{a^2+b^2-c^2}{2ab}$$

or, $\cos 60° = \frac{2^2+(\sqrt{3}+1)^2-c^2}{2\times 2\times(\sqrt{3}+1)}$

or, $\frac{1}{\cancel{2}} = \frac{4+3+2\sqrt{3}-c^2+1}{2\cancel{4}(\sqrt{3}+1)}$

or, $2(\sqrt{3}+1) = 8+2\sqrt{3}-c^2$

or, $c^2 = 8+\cancel{2\sqrt{3}}-\cancel{2\sqrt{3}}-2$

or, $c^2 = 6$

$\therefore c = \sqrt{6}$

Again,

$$\cos A = \frac{b^2+c^2-a^2}{2bc}$$

$$= \frac{(\sqrt{3}+1)^2+(\sqrt{6})^2-2^2}{2\times(\sqrt{3}+1)\times\sqrt{6}}$$

$$= \frac{3+2\sqrt{3}+1+6-4}{2\sqrt{6}(\sqrt{3}+1)}$$

$$= \frac{6+2\sqrt{3}}{2\sqrt{6}(\sqrt{3}+1)}$$

$$= \frac{2\sqrt{3}\,\cancel{(\sqrt{3}+1)}}{\cancel{2}\sqrt{6}\,\cancel{(\sqrt{3}+1)}}$$

$$= \frac{1}{\sqrt{2}}$$

$\therefore A = 45°$

Now,

we know that,

$A + B + C = 180°$

or, $45° + B + 60° = 180°$

or, $B = 75°$

$\therefore B = 75°$ //.

(10) In a $\triangle ABC$, if $a = \sqrt{57}$, $A = 60°$ and $\Delta = 2\sqrt{3}$, then find the measures of sides b and c.

→ Soln,

we have,

$$\Delta = \frac{1}{2} bc \sin A$$

or, $2\sqrt{3} = \frac{1}{2} \times bc \times \sin 60°$

or, $2\cancel{\sqrt{3}} = \frac{1}{2} \times bc \times \frac{\cancel{\sqrt{3}}}{2}$

or, $8 = bc$

or, $b = \frac{8}{c} \longrightarrow (i)$

Again,

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

or, $\cos 60 = \frac{b^2 + c^2 - 57}{2bc}$

or, $\frac{1}{\cancel{2}} = \frac{b^2 + c^2 - 57}{\cancel{2} \times 8}$

or, $8 + 57 = b^2 + c^2$

or, $b^2 + c^2 = 65 \longrightarrow (ii)$

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From ① & ②

$$\left(\frac{8}{c}\right)^2 + c^2 = 65$$

or, $\frac{64}{c^2} + c^2 = 65$

or, $64 + c^4 = 65c^2$

or, $c^4 - 65c^2 + 64 = 0$

or, $c^4 - 64c^2 - c^2 + 64 = 0$

or, $c^2(c^2 - 64) - 1(c^2 - 64) = 0$

or, $(c^2 - 1)(c^2 - 64) = 0$

Either, OR

$c^2 - 1 = 0$ $c^2 - 64 = 0$

$c^2 = 1$ or, $c^2 = 64$

$c = 1$ $\therefore c = 8$

When $c = 1$

$b = \frac{8}{c} = \frac{8}{1} = 8$

When $c = 8$

$b = \frac{8}{c} = \frac{8}{8} = 1$

$\therefore b = 8, c = 1$ or $b = 1, c = 8$

7a) $a = 2$, $b = \sqrt{2}$ and $c = \sqrt{3} + 1$

↳ Using cosine law,

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

$$= \frac{(\sqrt{2})^2 + (\sqrt{3}+1)^2 - 2^2}{2 \times \sqrt{2} \times (\sqrt{3}+1)}$$

$$= \frac{2 + 3 + 2\sqrt{3} + 1 - 4}{2\sqrt{2}(\sqrt{3}+1)}$$

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$$= \frac{\cancel{2}\cancel{(1+\sqrt{3})}}{\cancel{2}\sqrt{2}\cancel{(1+\sqrt{3})}}$$

$$= \frac{1}{\sqrt{2}}$$

$\therefore A = 45°$

Again,

$$\cos B = \frac{a^2+c^2-b^2}{2ac}$$

$$= \frac{2^2+(\sqrt{3}+1)^2-(\sqrt{2})^2}{2\times 2(\sqrt{3}+1)}$$

$$= \frac{4+3+2\sqrt{3}+1-2}{4(\sqrt{3}+1)}$$

$$= \frac{6+2\sqrt{3}}{4(\sqrt{3}+1)}$$

$$= \frac{\cancel{2}\sqrt{3}\cancel{(\sqrt{3}+1)}}{2\cancel{4}\cancel{(\sqrt{3}+1)}}$$

$$= \frac{\sqrt{3}}{2}$$

$\therefore B = 30°$

Now,

$A+B+C = 180°$

or, $45° + 30° + C = 180°$

or, $C = 180° - 75°$

$\therefore C = 105°$ #

8. If $a=2$, $b=\sqrt{6}$ and $c=\sqrt{3}+1$, find the greatest and the smallest angle of the ΔABC.

→ Soln,

Using cosine law,

$$\cos A = \frac{b^2+c^2-a^2}{2bc}$$

$$= \frac{(\sqrt{6})^2+(\sqrt{3}+1)^2-2^2}{2\times\sqrt{6}(\sqrt{3}+1)}$$

$$= \frac{6+3+2\sqrt{3}+1-4}{2\sqrt{6}(\sqrt{3}+1)}$$

$$= \frac{6+2\sqrt{3}}{2\sqrt{6}(\sqrt{3}+1)}$$

$$= \frac{\cancel{2}\sqrt{3}\cancel{(\sqrt{3}+1)}}{\cancel{2}\sqrt{6}\cancel{(\sqrt{3}+1)}}$$

$$= \frac{1}{\sqrt{2}}$$

$\therefore A = 45^\circ$

$$\cos B = \frac{a^2+c^2-b^2}{2ac}$$

$$= \frac{2^2+(\sqrt{3}+1)^2-(\sqrt{6})^2}{2\times 2\times(\sqrt{3}+1)}$$

$$= \frac{4+3+2\sqrt{3}+1-6}{4(\sqrt{3}+1)}$$

$$= \frac{2+2\sqrt{3}}{4(\sqrt{3}+1)}$$

$$= \frac{2\cancel{(1+\sqrt{3})}}{4\cancel{(1+\sqrt{3})}}$$

$$= \frac{1}{2}$$

$B = 60^\circ$

Now,

$A+B+C = 180^\circ$

$45^\circ + 60^\circ + C = 180^\circ$

or, $C = 180^\circ - 105^\circ$

$\therefore C = 75^\circ$

$\therefore$ Greatest angle $C = 75^\circ$

Smallest angle $A = 45^\circ$ #.

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9. b) $a=\sqrt{6}$, $c=2$, $C=45°$

↳ Using cosine rule,

$$\cos C = \frac{a^2+b^2-c^2}{2ab}$$

or, $\cos 45 = \frac{(\sqrt{6})^2+b^2-2^2}{2\times\sqrt{6}\times b}$

or, $\frac{1}{\sqrt{2}} = \frac{6+b^2-4}{2\sqrt{6}\,b}$

or, $\frac{1}{\sqrt{2}} = \frac{2+b^2}{2\sqrt{6}\,b}$

or, $b^2+2 = \sqrt{12}\,b$

or, $b^2-\sqrt{12}\,b+2=0$

or, $b^2-2\sqrt{3}\,b+2=0$

~~or, $b^2-\sqrt{3}b-b+b-\sqrt{3}b+2=0$~~

or, $b^2-2\sqrt{3}\,b+(\sqrt{3})^2-1=0$

or, $b^2-(\sqrt{3}-1)b-(1+\sqrt{3})b+(\sqrt{3}+1)(\sqrt{3}-1)=0$

or, $b[b-(\sqrt{3}-1)]-(1+\sqrt{3})[b-(\sqrt{3}-1)]=0$

or, $[b-(1+\sqrt{3})][b-(\sqrt{3}-1)]=0$

Either, $b-(1+\sqrt{3})=0$, $b=1+\sqrt{3}$

OR $b-(\sqrt{3}-1)=0$, $b=\sqrt{3}-1$

Using sine law,

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$\frac{\sqrt{6}}{\sin A} = \frac{b}{\sin B} = \frac{2}{\sin 45}$$

When $b=1+\sqrt{3}$

$$\frac{1+\sqrt{3}}{\sin B} = 2\sqrt{2}$$

or, $\sin B = \frac{1+\sqrt{3}}{2\sqrt{3}}$

$\therefore B = 75°, 105°$ (Not valid).

$$\frac{\sqrt{6}}{\sin A} = 2\sqrt{2}$$

or, $\sin A = \dfrac{\sqrt{6}}{2\sqrt{2}}$

$\sin A = \dfrac{\sqrt{3}}{2}$

$A = 60^\circ$ or 120° (Not valid)

When $b = \sqrt{3} - 1$

$$\frac{\sqrt{3}-1}{\sin B} = 2\sqrt{2}$$

or, $\sin B = \dfrac{\sqrt{3}-1}{2\sqrt{2}}$

$\therefore B = 15^\circ$

And,

$A + B + C = 180^\circ$

or, $45^\circ + 15^\circ + A = 180^\circ$

or, $A = 180^\circ - 60^\circ$

$\therefore A = 120^\circ$

$\therefore$ $A = 60^\circ$, $B = 75^\circ$, $b = \sqrt{3} + 1$ or.

$A = 120^\circ$, $B = 15^\circ$, $b = \sqrt{3} - 1$ #.